1.1 Exploratory Data Analysis (EDA)

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Data from 1960 to 2024 on Total Live Births (TLB) and Total Fertility Rates (TFR) was pulled from Singapore Department of Statistics (data.gov.sg). The data was then transformed into a tsibble object with three columns: Year, Total Fertility Rates (TFR) and Total Live Births.

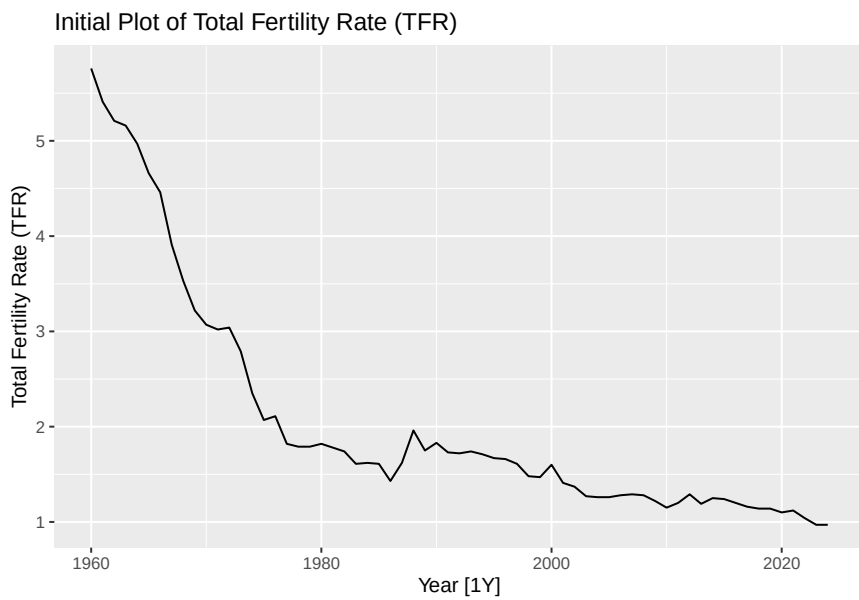
```r
library(fpp3) # time series analysis package
library(tsibble) # tsibble objects
library(lubridate) # for data to work with dates and times
library(feasts)
library(latex2exp)
library(patchwork) # putting 2 plots next to each other

data <- readr::read_csv("BirthsAndFertilityRatesAnnual.csv") # read in the data
BirthAndFertility <- data |>
  pivot_longer( # change rows to column
    cols = `1960`:`2025`,
    names_to = "Year",
    values_to = "Value", # naming it to remember that it needs to be values
    values_transform = list(Value = as.numeric)
  ) |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births")) |>
  mutate(Year = as.integer(Year)) |>
  filter(Year != 2025) |> # assignment specified until 2024
  pivot_wider( # taking TFR & Total Live-Births into 2 separate columns according to year
  names_from = DataSeries,
  values_from = Value
  )
BirthAndFertility <- as_tsibble(BirthAndFertility, index = Year) # created a tsibble
is_tsibble(BirthAndFertility) # checking it is a tsibble
```

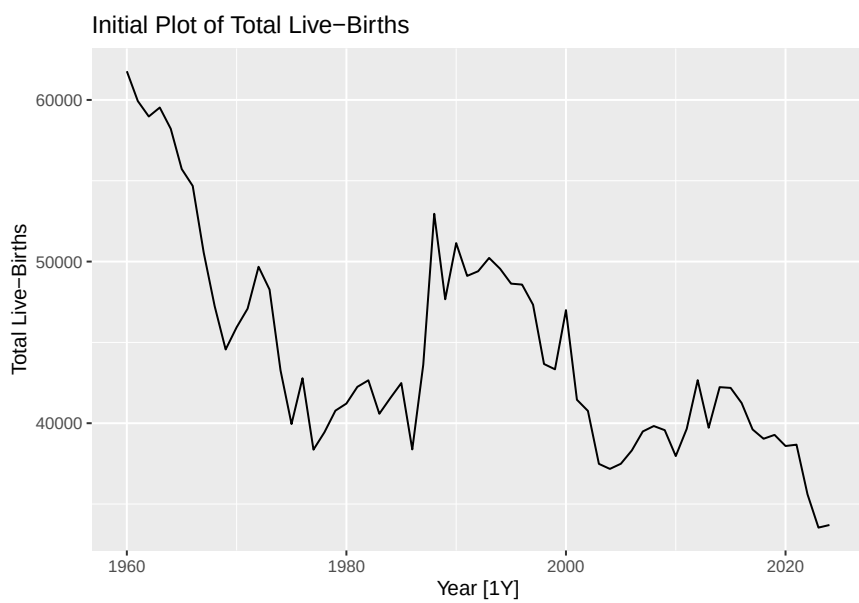
```
## [1] TRUE
```

In this exploratory data analysis, the data was visualised to identify patterns and trends over time. Since the data is yearly, classical decomposition and the X-11 method are not applicable. Once the patterns are identified, appropriate models can be selected, fitted, and evaluated for forecasting.

```r
autoplot(BirthAndFertility, `Total Fertility Rate (TFR)`) +
  labs(title = "Initial Plot of Total Fertility Rate (TFR)")
```



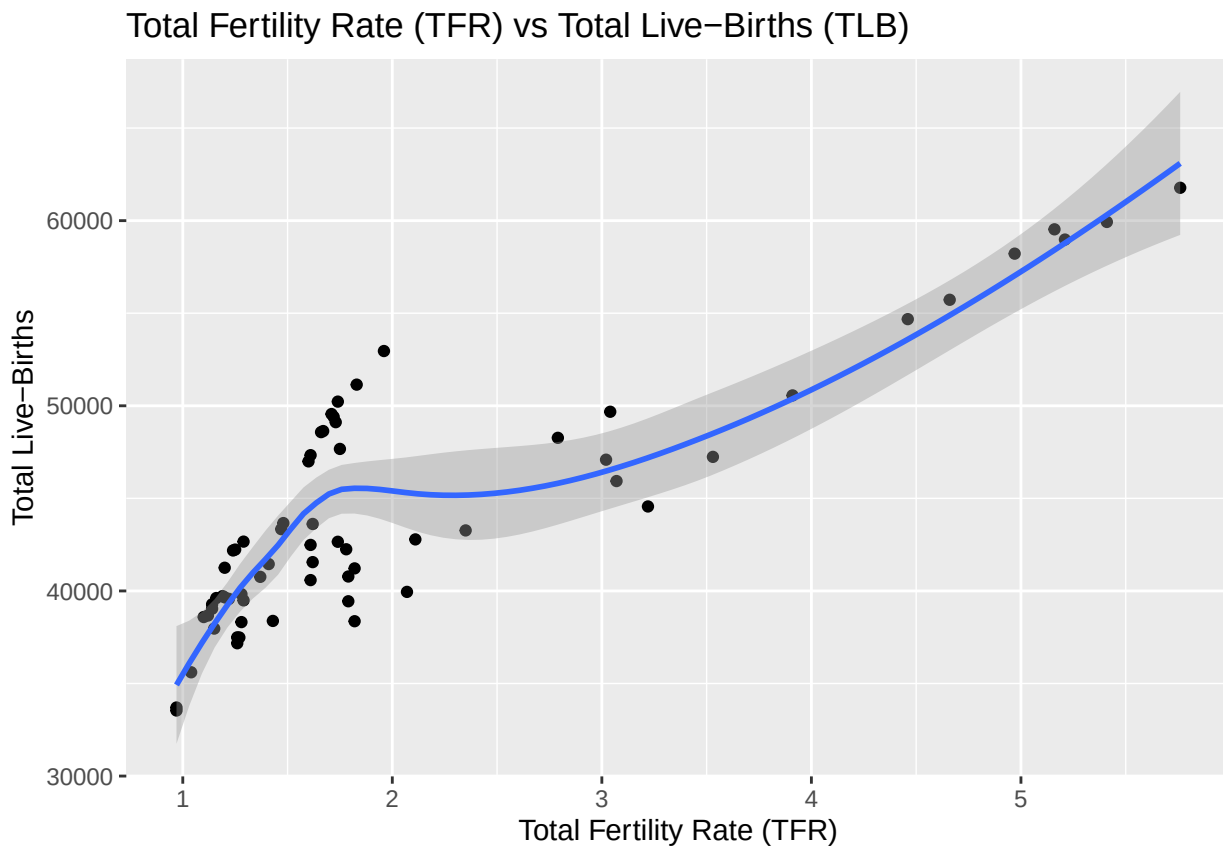

```r
autoplot(BirthAndFertility,`Total Live-Births`)+
  labs(title = "Initial Plot of Total Live-Births")
```



From the initial plots of TFR and TLB, both series show a decreasing trend over time, with TFR exhibiting a more pronounced decline. As the data is yearly, seasonality cannot be determined. While both series may suggest cyclic behaviour, the trend would need to be removed to confirm this.

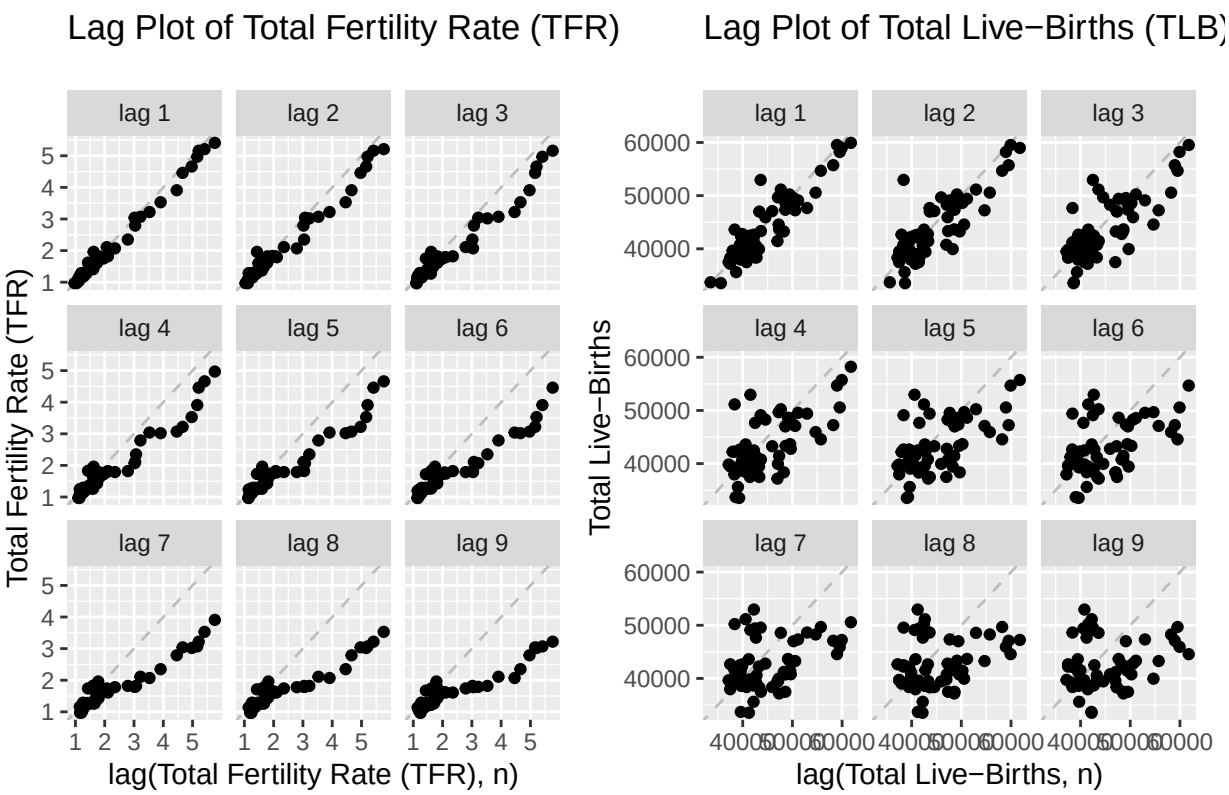
```r
BirthAndFertility |>
  ggplot(aes(x = `Total Fertility Rate (TFR)`, y = `Total Live-Births`)) +
  geom_point() +
  geom_smooth() +
  labs(title = "Total Fertility Rate (TFR) vs Total Live-Births (TLB)")
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



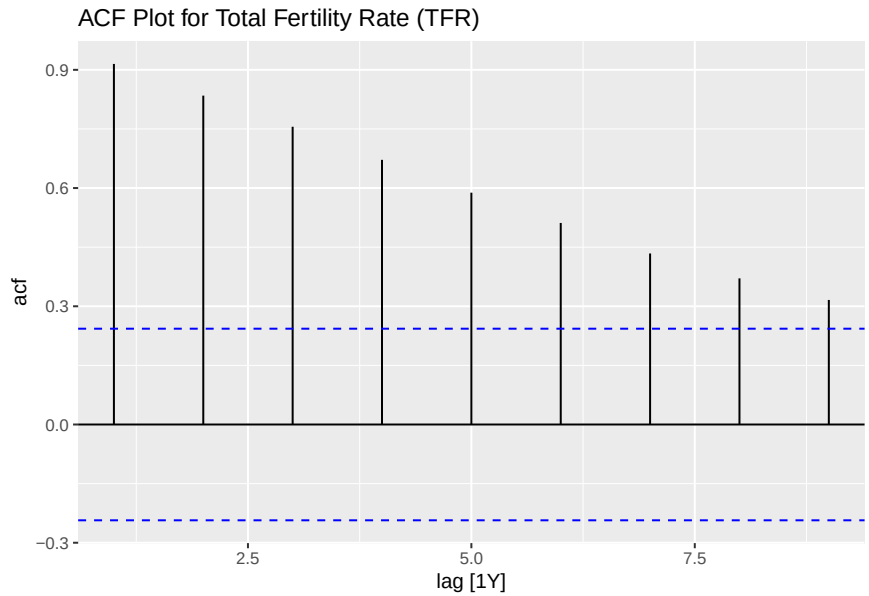
The relationship between TFR and TLB was examined by plotting the two variables against each other. There appears to be a nonlinear positive relationship, suggesting that higher fertility rates are associated with higher numbers of live births, although further analysis is required.

```r
p1 <- BirthAndFertility |>
  gg_lag(`Total Fertility Rate (TFR)`, geom = 'point') +
  labs(title = "Lag Plot of Total Fertility Rate (TFR)")
p2 <- BirthAndFertility |>
  gg_lag(`Total Live-Births`, geom = 'point') +
  labs(title = "Lag Plot of Total Live-Births (TLB)")
p1 + p2 # placing the plots side by side
```



The above lag plots for TFR and TLB indicate stronger positive relationships at lags 1 and 2 suggesting that AR(1) and AR(2) models could be appropriate.

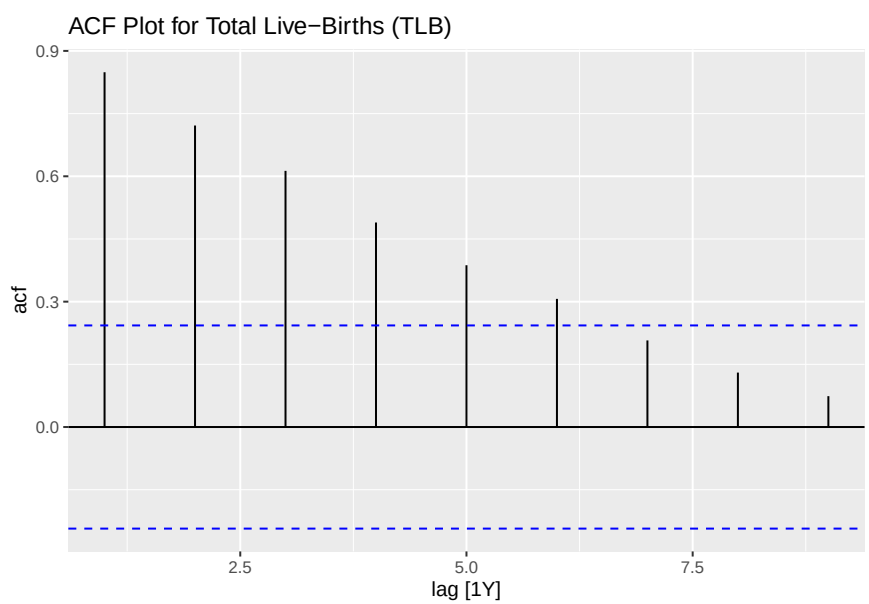
```r
BirthAndFertility |>
  ACF(`Total Fertility Rate (TFR)`, lag_max = 9) |>
  autoplot() +
  labs(title = "ACF Plot for Total Fertility Rate (TFR)")
```



lags 1-9 are significant so more lags should be checked

slow decrease in the ACF as the lags increase indicates a trend

```r
BirthAndFertility |>
  ACF(`Total Live-Births`, lag_max = 9) |>
  ggplot2::autoplot() +
  labs(title = "ACF Plot for Total Live-Births (TLB)")
```

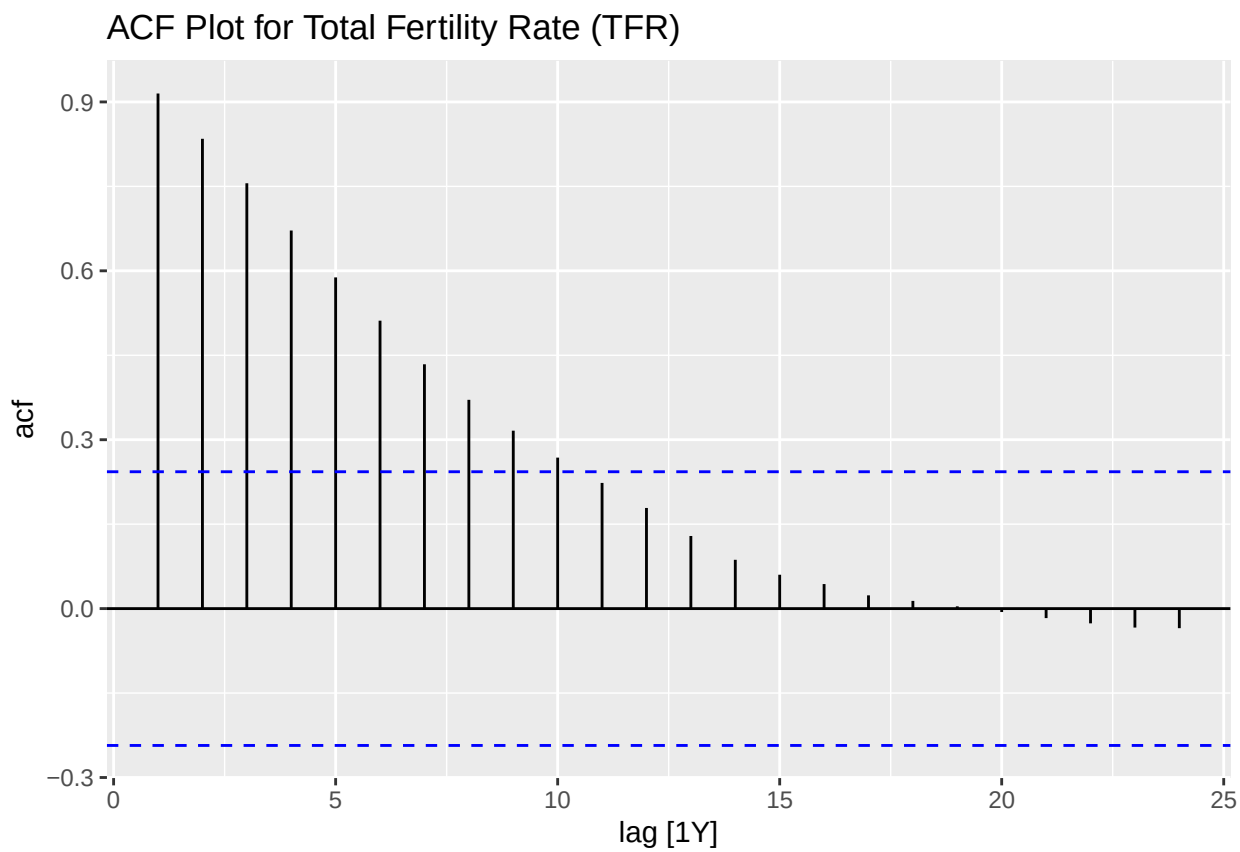


lags 1-6 are significant suggests short-term dependence

slow decrease in the ACF as the lags increase indicates a trend

as there are more than one large spike outside the dotted bounds, this series is probably not white noise

```
BirthAndFertility |>
  ACF(`Total Fertility Rate (TFR)`, lag_max = 24) |>
  autoplot() +
  labs(title = "ACF Plot for Total Fertility Rate (TFR)")
```



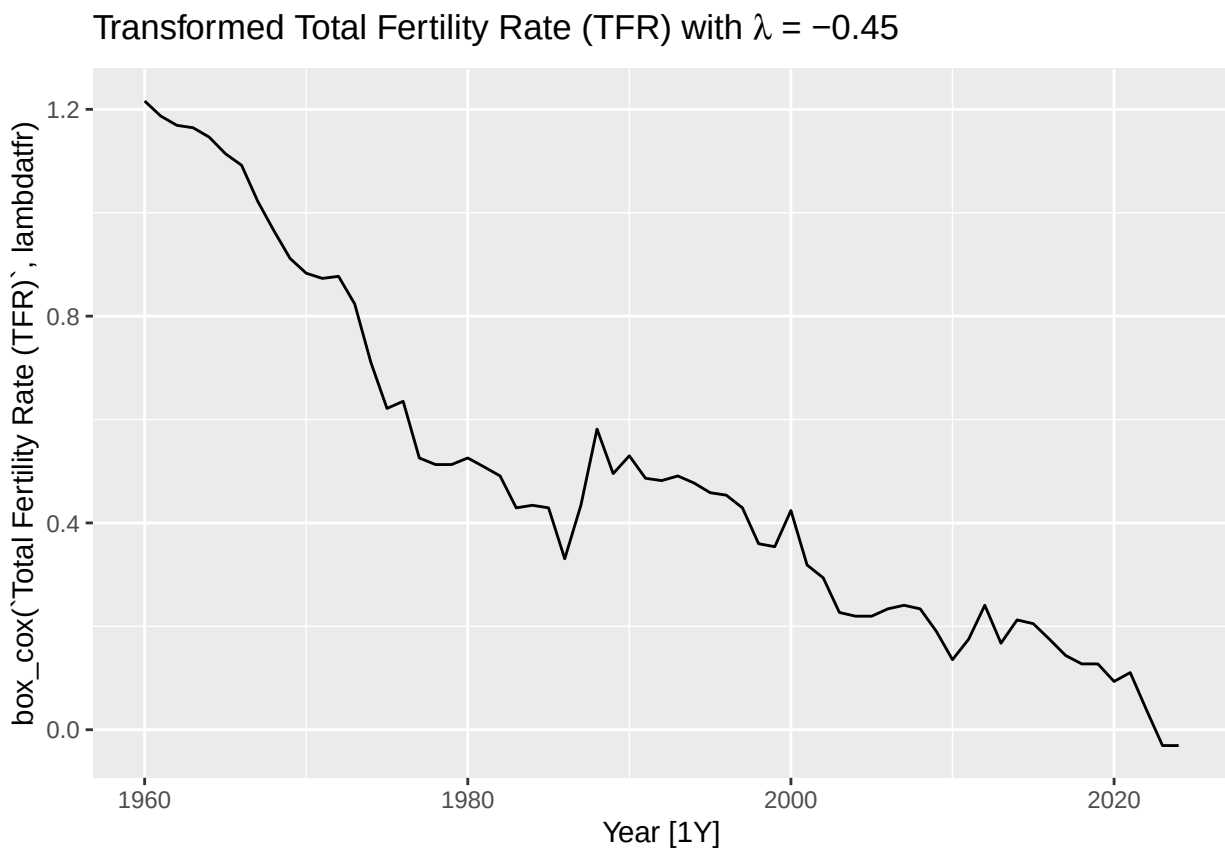
lags 1-10 are significant suggests strong persistence # slow decrease in the ACF as the lags increase indicates a trend # as there are more than one large spike outside the dotted bounds, this series is probably not white noise

Box-Cox transformation

```
lambdatfr <- BirthAndFertility |>
  features(`Total Fertility Rate (TFR)`, features = guerrero) |> # Guerrero method
  pull(lambda_guerrero) # Box-Cox transformation parameter = -0.44514
```

as lambda is below zero but halfway to -1 suggests moderate transformation required

```
BirthAndFertility |>
  autoplot(box_cox(`Total Fertility Rate (TFR)`, lambdatfr)) +
  labs(title = latex2exp::TeX(paste0(
    "Transformed Total Fertility Rate (TFR) with $\\lambda$ = ", round(lambdatfr,2))))
```

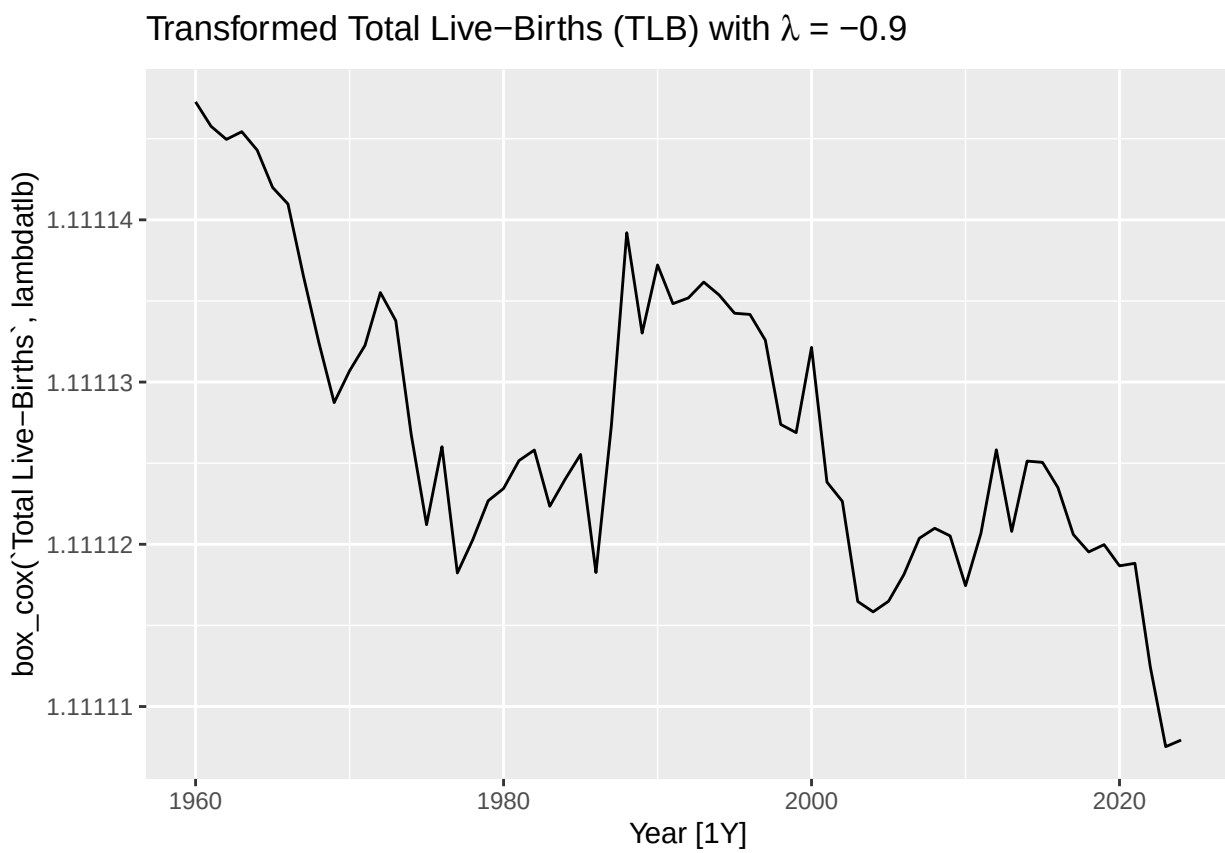


decreasing trend still obvious # still signs there might also be cyclic but still will have to take away trend to find out

```
lambdatlb <- BirthAndFertility |>
  features(`Total Live-Births`, features = guerrero) |> # Guerrero method
  pull(lambda_guerrero) # Box-Cox transformation parameter = -0.89992
```

as lambda is below zero but close to -1 suggests strong transformation required

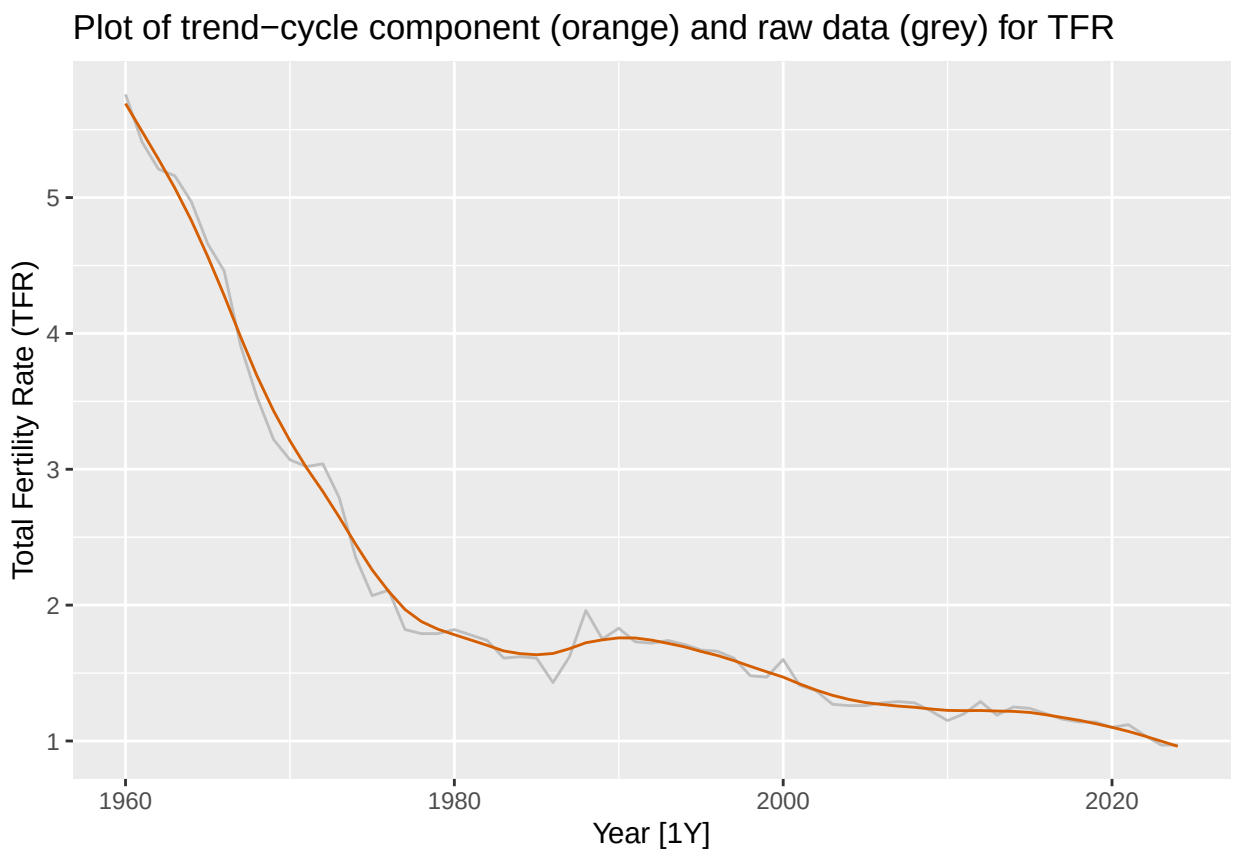
```
BirthAndFertility |>
  autoplot(box_cox(`Total Live-Births`, lambdatlb)) +
  labs(title = latex2exp::TeX(paste0(
    "Transformed Total Live-Births (TLB) with $\\lambda$ = ", round(lambdatlb,2))))
```



the decreasing trend from 1960 to 2024 is less prominent # might still be cyclic but still will have to take away trend to find out

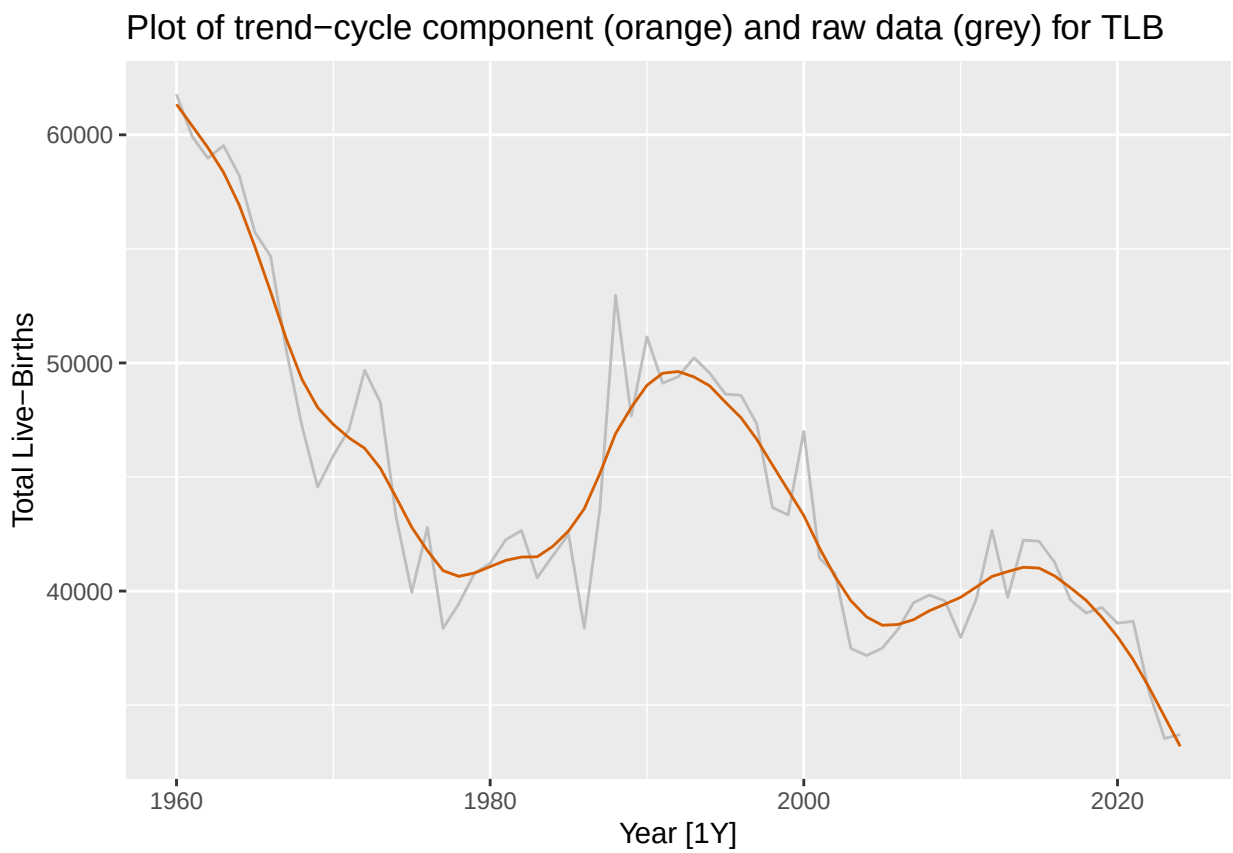
STL decomposition method: trend

```
dcmptfr <- BirthAndFertility |>
  model(stl = STL(`Total Fertility Rate (TFR)`))
components(dcmptfr) |>
  as_tsibble() |>
  autoplot(`Total Fertility Rate (TFR)`, colour = "grey") + # raw data
  geom_line(aes(y = trend), colour = "#D55E00") + # trend-cycle component
  labs(title = "Plot of trend-cycle component (orange) and raw data (grey) for TFR")
```



little variability to the raw data, more smooth

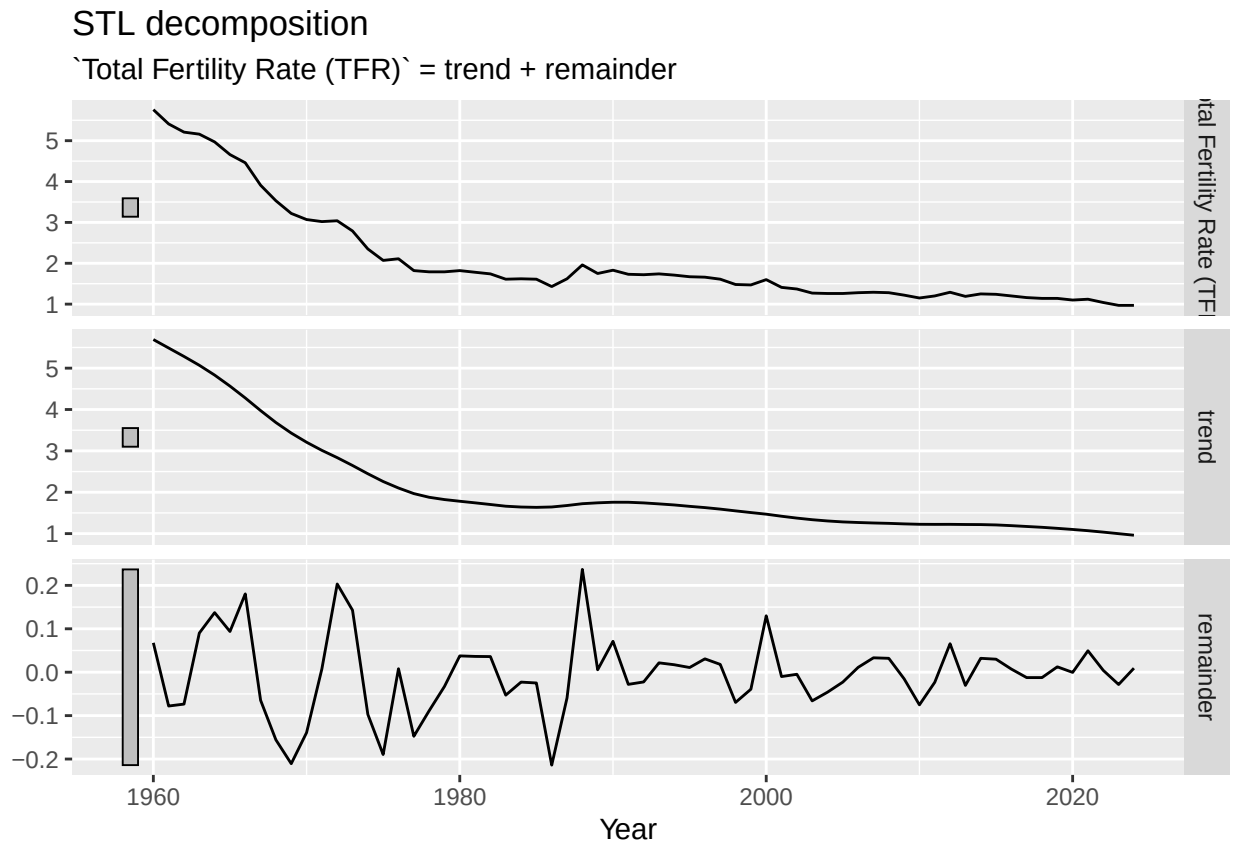
```
dcmptlb <- BirthAndFertility |>
  model(stl = STL(`Total Live-Births`))
components(dcmptlb) |>
  as_tsibble() |>
  autoplot(`Total Live-Births`, colour = "grey") + # raw data
  geom_line(aes(y = trend), colour = "#D55E00") + # trend-cycle component
  labs(title = "Plot of trend-cycle component (orange) and raw data (grey) for TLB")
```



more variability to the raw data, bigger changes

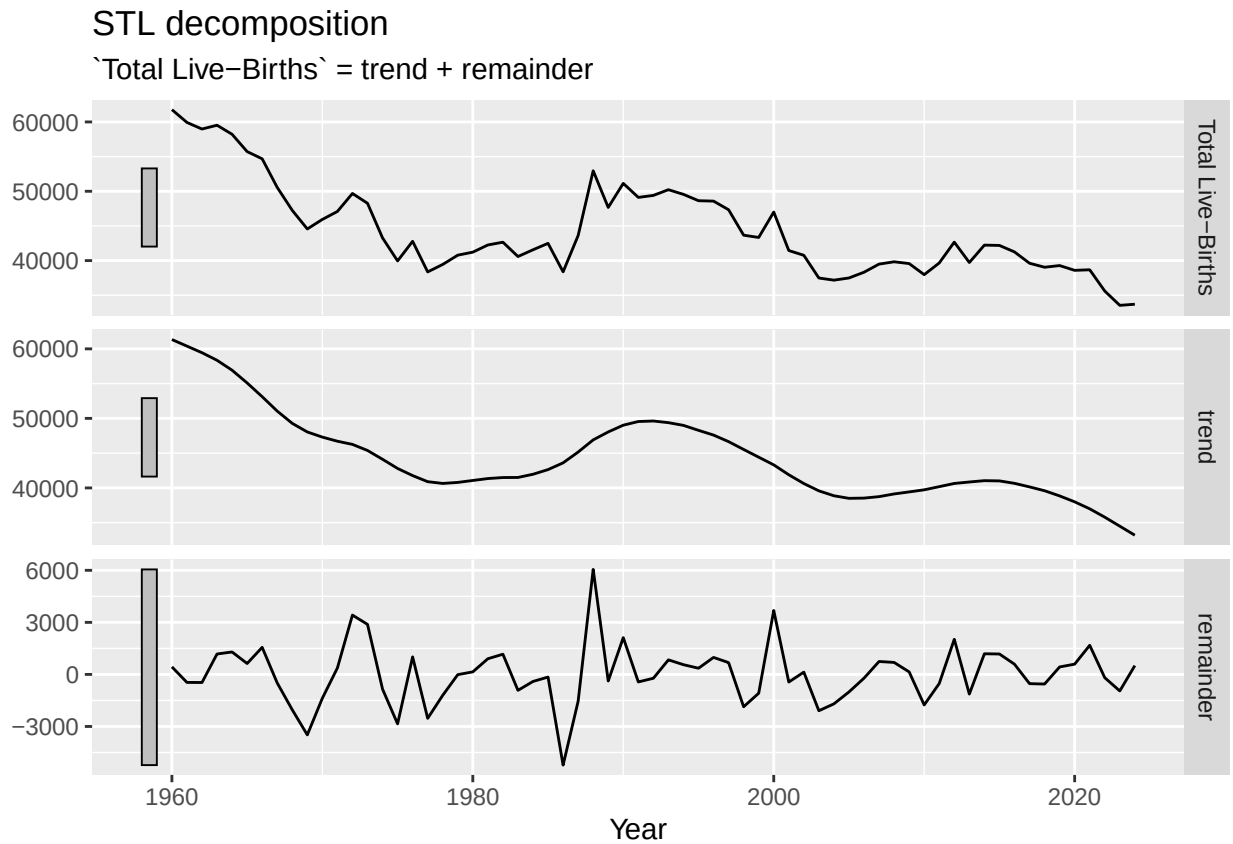
STL decomposition

```
components(dcmptfr) |> autoplot()
```



decreasing trend as mentioned earlier

```
components(dcmptlb) |> autoplot()
```



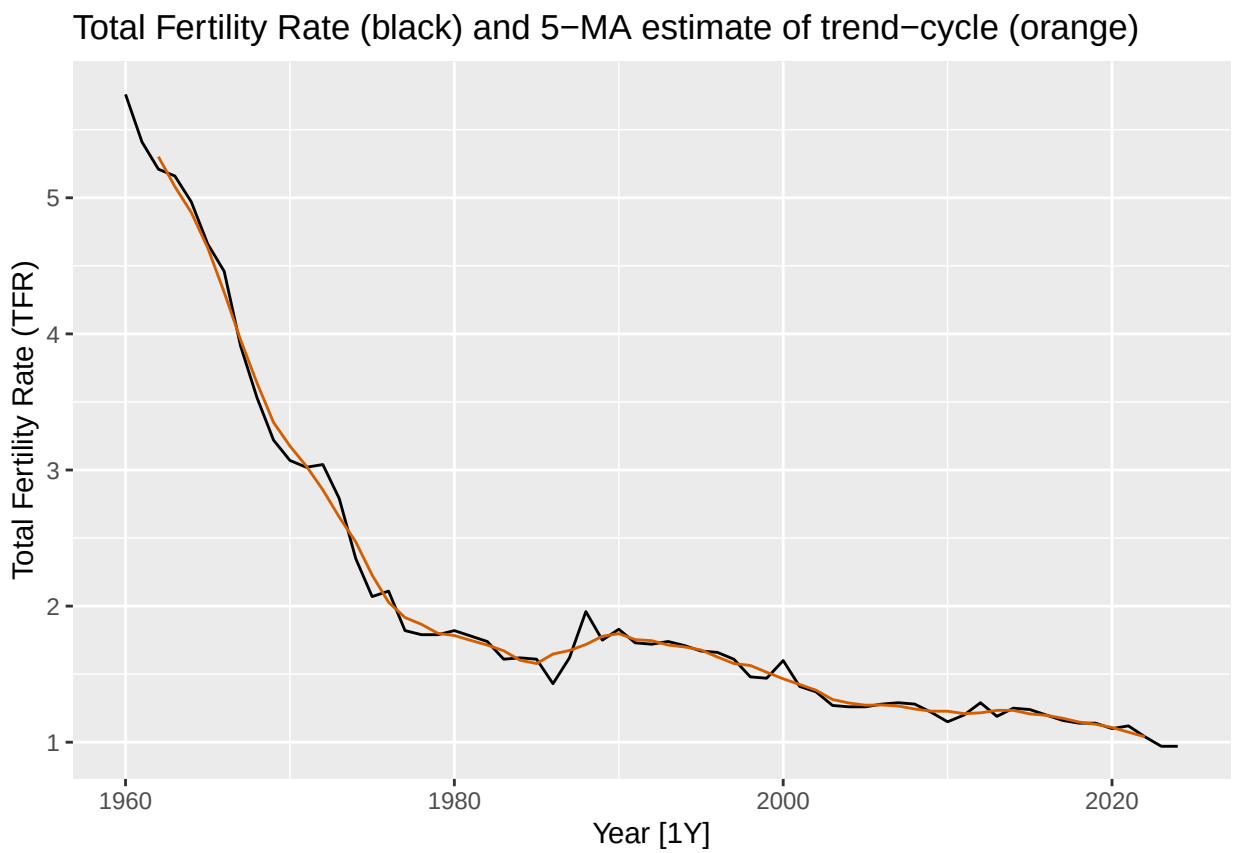
overall decreasing trend but there was an increase from 1980 with a peak in 1991 before decreasing again

the remainder of both decompositions resemble each other with sharp dip around 1986 and spike 1988

moving average

```
BirthAndFertility_MA_TFR <- BirthAndFertility |>
  mutate(`5-MA_TFR` = slider::slide_dbl(`Total Fertility Rate (TFR)`, mean,
                                  .before = 2, .after = 2, .complete = TRUE))
BirthAndFertility_MA_TFR |>
  autoplot(`Total Fertility Rate (TFR)`) +
  geom_line(aes(y = `5-MA_TFR`), colour = "#D55E00") +
  labs(title = "Total Fertility Rate (black) and 5-MA estimate of trend-cycle (orange)")
```

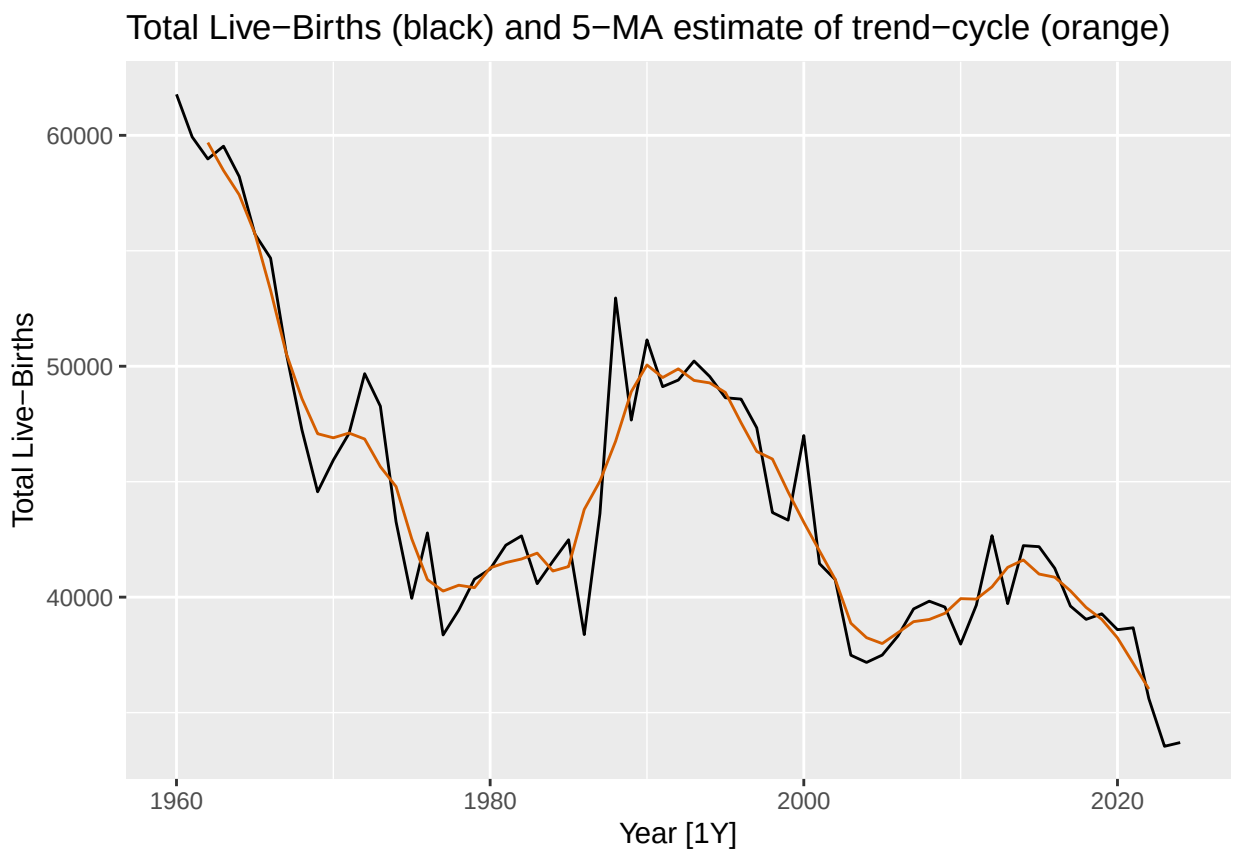
```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



the original data was already quite smooth so it matches quite closely with the trend-cycle

```
BirthAndFertility_MA_TLB <- BirthAndFertility |>
  mutate(`5-MA_TLB` = slider::slide_dbl(`Total Live-Births`, mean,
                                        .before = 2, .after = 2, .complete = TRUE))
BirthAndFertility_MA_TLB |>
  autoplot(`Total Live-Births`) +
  geom_line(aes(y = `5-MA_TLB`), colour = "#D55E00") +
  labs(title = "Total Live-Births (black) and 5-MA estimate of trend-cycle (orange)")
```

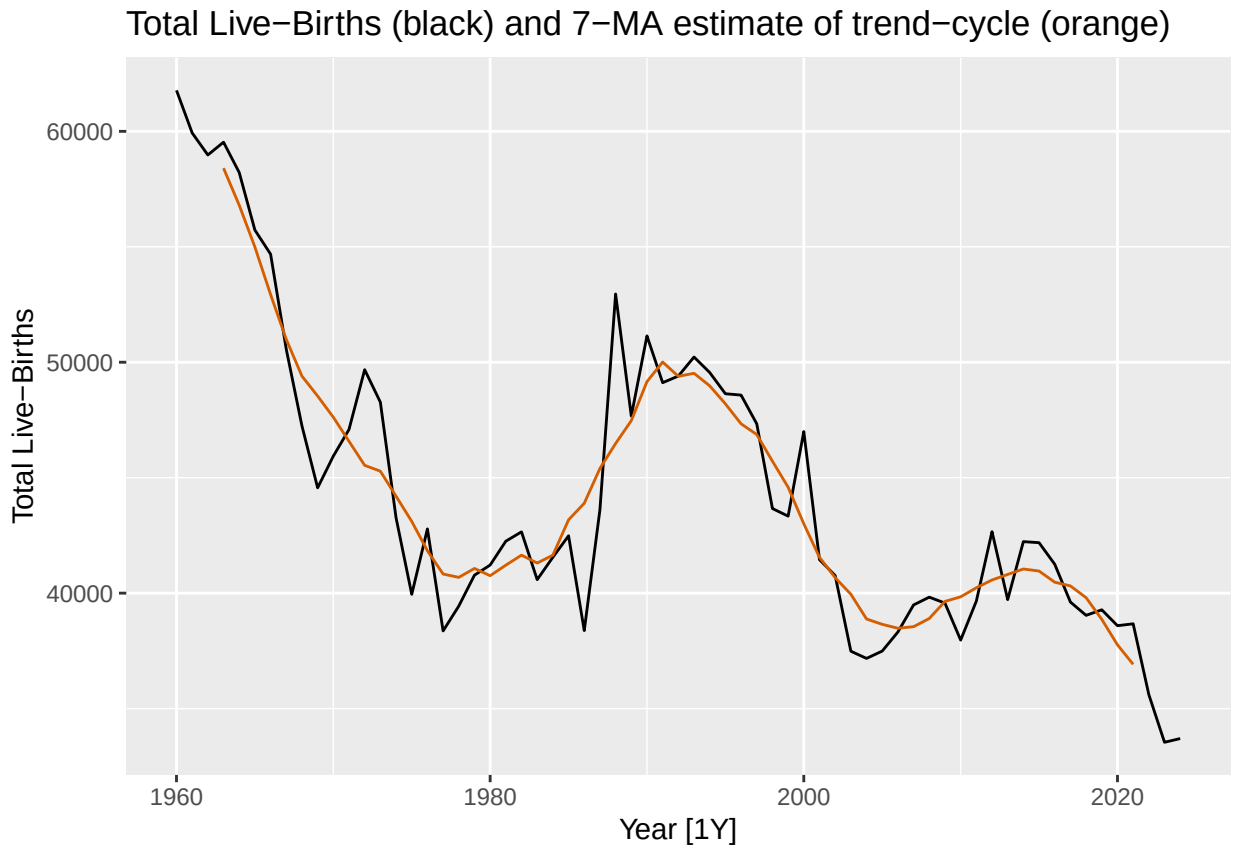
```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



can see smooth curve but due to the roughness of the data should increase the average for a smoother curve

```
BirthAndFertility_MA_TLB7 <- BirthAndFertility |>
  mutate(`7-MA_TLB` = slider::slide_dbl(`Total Live-Births`, mean,
                                        .before = 3, .after = 3, .complete = TRUE))
BirthAndFertility_MA_TLB7 |>
  autoplot(`Total Live-Births`) +
  geom_line(aes(y = `7-MA_TLB`), colour = "#D55E00") +
  labs(title = "Total Live-Births (black) and 7-MA estimate of trend-cycle (orange)")
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



the curve is smoother at the cost of the ends

evidence of the presence of business cycles?

outliers that need to be explained by experts?